

Chapter 1 Notes: Subject Matter Overview (Automatic Control Systems)

1. The Big Picture: What Is a Control System?

A **control system** is a set of devices that change the *natural* behavior of a **plant** (the thing you're trying to control) into a *desired* behavior.

Think of it in two layers:

Plant — the physical thing being controlled (a rocket, an oven, a bicycle).

Control system — the sensors + effectors that watch the plant and push it toward the behavior you want.

Three worked examples

System	Plant	Control System (sensors/actuators)	Natural (uncontrolled) Behavior	Controlled Behavior
Rocket during launch	Rocket (mechanical, electrical, electronic, chemical)	Sensors measure position/attitude; actuators steer the engine nozzle	Thrust + gravity drive the rocket into the ground	Thrust is aimed to propel the rocket into space
Home oven	Oven (electronic + chemical/fuel systems)	Thermostat + fuel-line flow control	Heats to max temperature possible	Heats to a set temperature and holds it
Self- balancing bicycle	Bicycle (mechanical)	Sensors measure tip angle; actuator steers the front wheel	Bicycle falls over	Bicycle stays balanced while moving forward

Key intuition: without control, these systems drift toward a bad outcome (crash, overheating, falling). The control system's job is to redirect that natural tendency into something useful and safe.

Stability — a first look

If an uncontrolled system's response grows without bound and damages the plant, it's called **unstable** (a precise math definition comes later in the course).

Example: rocket without guidance → crashes. Bicycle without balancing → falls. Both are "unstable" in this sense.

2. Three Areas of Control Systems Study

Model development — finding a mathematical model that represents the real system.

Analysis — studying a system with a known/proven model using established techniques.

Controller design — designing and building a controller to make the system behave as desired.

This course focuses mainly on **analysis** and **controller design**, using **linear, time-invariant (LTI)** models (the "nice," well-behaved case). Nonlinear/time-varying systems need more advanced tools that are beyond this course's scope. In practice, engineers often just simulate systems in software (e.g., MATLAB) to see the time-domain behavior directly.

3. Why We Use the Laplace Domain

Working directly with differential equations (time domain) is often messy. Instead, we convert to the **Laplace domain**, where dynamic behavior can be studied algebraically instead of by solving differential equations.

Time domain ↔ Laplace domain conversions are done with a **Look-Up Table (LUT)** — you are not expected to derive transforms from the defining integral each time.

The two directions are related by the Laplace transform pair:

$$F(s) = \int_0^{\infty} f(t)e^{-st} dt$$

$$f(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} F(s)e^{st} ds$$

You'll rarely use these formulas directly — the LUT (Section 1.2/1.3 table) does the work.

Worked example: converting a differential equation to a transfer function

Start with a time-domain ODE:

$$\ddot{y}(t) + a_1\dot{y}(t) + a_0y(t) - \dot{u}(t) - b_0u(t) = 0$$

Step 1 — Laplace transform each term (using the derivative rules from the LUT):

$$s^2Y(s) - sy(0) - \dot{y}(0) + a_1sY(s) - a_1y(0) + a_0Y(s) - sU(s) + u(0) - b_0U(s) = 0$$

Step 2 — Assume zero initial conditions (standard assumption in this course, i.e. $y(0) = \dot{y}(0) = u(0) = 0$):

$$s^2Y(s) + a_1sY(s) + a_0Y(s) - sU(s) - b_0U(s) = 0$$

Step 3 — Group inputs on one side, outputs on the other:

$$s^2Y(s) + a_1sY(s) + a_0Y(s) = sU(s) + b_0U(s)$$

Step 4 — Factor out Y(s) and U(s):

$$(s^2 + a_1s + a_0)Y(s) = (s + b_0)U(s)$$

Step 5 — Divide to isolate the ratio Y(s)/U(s):

$$\frac{Y(s)}{U(s)} = \frac{s + b_0}{s^2 + a_1s + a_0}$$

This ratio is called a **transfer function**.

Transfer function = Output / Input (always in the Laplace domain). In this course, output is always $y(t)/Y(s)$; input is either $u(t)/U(s)$ or $r(t)/R(s)$.

Why this matters: Real devices often come with a transfer function on their datasheet. You can cascade (chain together) transfer functions of individual components to build a model of a whole system — and analyze performance *before ever building the physical system*.

4. Basic Building-Block Models (Section 1.1)

(a) Amplifier

Input U: low-voltage/low-power signal

Output Y: higher-voltage/higher-power signal

Device: op-amp

Transfer function:

$$\frac{Y(s)}{U(s)} = K_{oa}, \quad -\infty < K_{oa} < \infty$$

Two op-amp wiring configurations give different gains:

Configuration	Gain K_{oa}
Non-inverting (input to $\boxed{+}$ terminal)	$K_{oa} = 1 + \frac{R_f}{R_i} = \frac{R_i + R_f}{R_i}$
Inverting (input to $\boxed{-}$ terminal)	$K_{oa} = -\frac{R_f}{R_i}$

(R_f = feedback resistor, R_i = input resistor.)

(b) Summation Junction

Input U_i : multiple signal sources

Output Y: the sum of those signals

Device: op-amp (summing amplifier)

$$Y(s) = \sum_{i=1}^n U_i(s)$$

⚠ **Note:** a summation junction is *not* a transfer function — it doesn't scale input to output by a ratio; it just adds signals (possibly with a gain applied to the sum, as below).

Example — 4-input summing amplifier, using the same R_f, R_i op-amp setup:

$$Y = \left(1 + \frac{R_f}{R_i}\right) (U_1 + U_2 + U_3 + U_4) = \left(\frac{R_i + R_f}{R_i}\right) (U_1 + U_2 + U_3 + U_4)$$

5. Laplace Transform Look-Up Table (Key Entries)

You don't need to memorize the derivation — just know how to use the table.

f(t)	F(s)	Notes
$k \cdot f(t)$	$kF(s)$	scaling
$f_1(t) + f_2(t)$	$F_1(s) + F_2(s)$	linearity
$\delta(t)$	1	impulse
$1(t)$	$1/s$	step
t	$1/s^2$	ramp
t^n	$n!/s^{n+1}$	
e^{-at}	$1/(s+a)$	exponential
$\sin(bt)$	$b/(s^2 + b^2)$	
$\cos(bt)$	$s/(s^2 + b^2)$	
$e^{-at} \sin(bt)$	$b/((s+a)^2 + b^2)$	
$e^{-at} \cos(bt)$	$(s+a)/((s+a)^2 + b^2)$	
$\dot{f}(t)$	$sF(s) - f(0)$	derivative; $f(0)$ = initial condition (assumed 0 in this course)
$\int f(t) dt$	$F(s)/s$	integration
$f_1(t) * f_2(t)$	$F_1(s) \cdot F_2(s)$	convolution in time $\leftrightarrow$ multiplication in s
$f_1(t) \cdot f_2(t)$	$F_1(s) * F_2(s)$	multiplication in time $\leftrightarrow$ convolution in s
$\lim_{t \rightarrow 0} f(t)$	$\lim_{s \rightarrow \infty} sF(s)$	Initial Value Theorem (IVT)
$\lim_{t \rightarrow \infty} f(t)$	$\lim_{s \rightarrow 0} sF(s)$	Final Value Theorem (FVT)

Tip: IVT and FVT let you find a signal's starting value or its steady-state (long-run) value directly from $F(s)$, without transforming back to $f(t)$.

6. Common Test/Input Signals (Section 1.3)

These are the standard signals used to "probe" a system and evaluate how well it tracks different kinds of inputs.

Signal	$f(t)$	Shape	Physical meaning / use
Impulse	$\delta(t)$	infinitely tall, zero width, area = 1	Not physically realizable, but useful in math/simulation.
Step	$f(t) = \begin{cases} 1, & t > 0 \\ 0, & t \leq 0 \end{cases}$	jumps from 0 to 1 and holds	A setpoint input — physically realizable and bounded (e.g., setting a thermostat to a target temperature).
Ramp	$f(t) = t$	straight line, increasing forever	Unbounded input; tests how well a system tracks a constant velocity .
Parabola	$f(t) = t^2$	curves upward, increasing forever	Unbounded input; tests how well a system tracks a constant acceleration .

Exponential — an *output*, not an input

Unlike the signals above, the exponential $f(t) = e^{-at}$ is treated as a typical **output** response of a transfer function, and its shape tells you about stability:

If $a > 0$: the response decays to 0 as $t \rightarrow \infty \rightarrow$ **stable**.

If $a < 0$: the response grows without bound as $t \rightarrow \infty \rightarrow$ **unstable**.

This is your first concrete link between a system's math (the sign of a constant in the exponent) and its real-world behavior (settles down vs. blows up) — a theme that gets developed much further later in the course.

Quick-Reference Summary

Control system = sensors + effectors that redirect a plant's natural (often bad) behavior into desired behavior.

Modeling workflow: time-domain ODE $\rightarrow$ Laplace transform (via LUT, zero initial conditions) $\rightarrow$ algebraic manipulation $\rightarrow$ **transfer function** $Y(s)/U(s)$.

Transfer function = output/input in the Laplace domain; can be cascaded to model complex systems.

Building blocks: amplifiers (scale a signal by gain K_{oa}) and summing junctions (add signals — not a transfer function).

Standard test inputs: impulse, step, ramp, parabola — increasing "difficulty" for a system to track.

Stability preview: exponential response e^{-at} decays (stable) if $a > 0$, grows (unstable) if $a < 0$.